

Random Sign Augmentation Beats Canonicalization for Laplacian Positional Encodings on ogbg-molpcba

ogbg-molpcba experiments, GIN backbone

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Abstract

On ogbg-molpcba with a 5-layer GIN and $k \in \{3, 8, 16\}$ Laplacian eigenvectors as positional encoding, random sign-flip augmentation (**random_augmented**) outperforms deterministic canonicalization (**map** and **oap**) at 8 of 9 tested (k, h) cells. The advantage grows with both k and h : at $k=16$, **random_augmented** beats **oap** by +0.0031 to +0.0086 test AP and **map** by +0.0081 to +0.0137.

1 Setup

Three approaches to the sign + basis ambiguity of LapPE:

- **map** — maximum-absolute-position canonicalization (deterministic).
- **oap** — ordered-absolute-position canonicalization (deterministic).
- **random_augmented** — training-time random sign flips per eigenvector per epoch; the network learns to be invariant to the sign ambiguity.

Hyperparameters. GIN, 5 layers, batch 32, lr 10^{-3} , Adam + cosine schedule. Hidden dim $h \in \{16, 128, 512\}$. Seeds $\{0, 1, 2\}$. Eigval-scaling off. Cache: 15 eigenvectors. PyTorch 2.10, CUDA 12.8, conda env **schnet-angle**.

2 Headline result

3 $k=16$ focus

The largest gaps are at $k=16$. Mean $\pm$ sample-std (ddof= 1), 3 seeds:

4 Bottom line

- **Random sign-flip augmentation outperforms canonicalization on ogbg-molpcba with GIN at every (k, h) tested except $k=3/h=512$.**
- The advantage grows with k : at $k=16$ the gap is +0.0031 to +0.0086 AP across $h \in \{16, 128, 512\}$ versus **oap**, and +0.0081 to +0.0137 versus **map**.
- **Practical recipe** for ogbg-molpcba-like data with LapPE: prefer **random_augmented** with $k \geq 8$ and hidden dim ≥ 128 .

k	h	map	oap	random_augmented	winner	Δ vs. best canon
3	16	0.0626	0.0641	0.0662	rand_aug	+0.0021
3	128	0.1704	0.1682	0.1762	rand_aug	+0.0058
3	512	0.2397	0.2431	0.2395	oap	-0.0036
8	16	0.0622	0.0605	0.0636	rand_aug	+0.0014
8	128	0.1705	0.1754	0.1766	rand_aug	+0.0012
8	512	0.2344	0.2371	0.2395	rand_aug	+0.0024
16	16	0.0610	0.0660	0.0691	rand_aug	+0.0031
16	128	0.1779	0.1834	0.1916	rand_aug	+0.0082
16	512	0.2381	0.2397	0.2483	rand_aug	+0.0086

Table 1: Mean test AP, 3 seeds per cell. **random_augmented** beats both canonicalization arms (**map**, **oap**) at 8 of 9 tested (k, h) cells. The exception is $k=3/h=512$, where **oap** edges **random_augmented** by 0.0036 AP. The Δ column reports the augmented-vs-best-canonicalization gap. At $k=16$ the gap widens monotonically with h .

h	map	oap	random_augmented
16	0.0610 ± 0.0009	0.0660 ± 0.0054	0.0691 ± 0.0008
128	0.1779 ± 0.0065	0.1834 ± 0.0036	0.1916 ± 0.0009
512	0.2381 ± 0.0147	0.2397 ± 0.0061	0.2483 ± 0.0026

Table 2: $k=16$ headline. **random_augmented** wins at every h by margins that exceed both seed std values. The trend is monotone in h : bigger model, bigger gap.

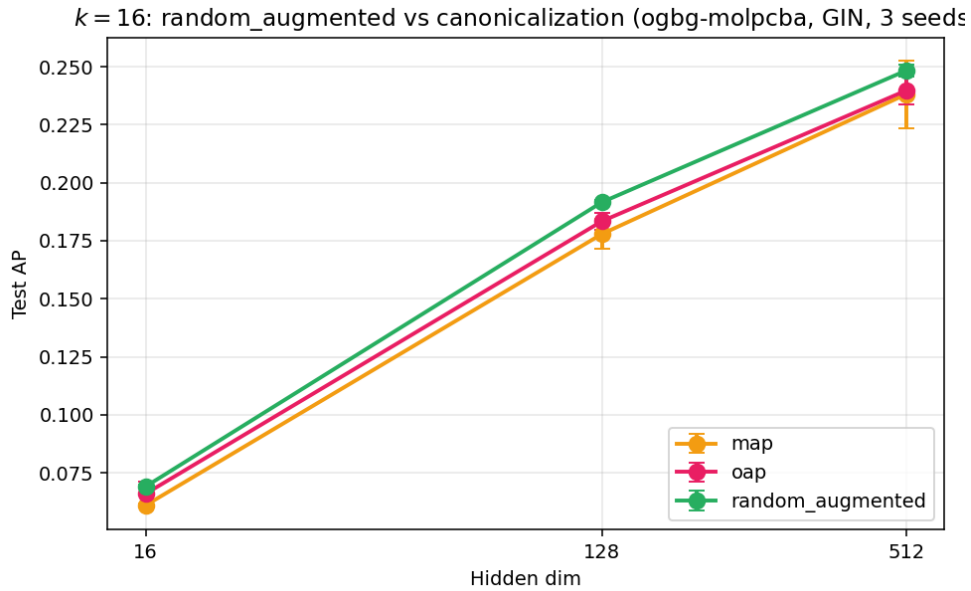


Figure 1: $k=16$ test AP vs. hidden dim, 3 seeds each. **random_augmented** (green) sits above both **oap** (pink) and **map** (orange) at every h .